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CLASS:P5

C PROGRAMS WEEK-6

PROGRAM-1

CODE

```c
#include<stdio.h>


float calculateHRA(float basic)
{return 0.2*basic;}

float calculateDA(float basic)
{return 0.1*basic;}

float calculateGross(float basic, float hra, float da)
{return basic+hra+da;}

float calculateTax(float gross)
{return 0.05*gross;}

float calculateNet(float gross,float tax)
{return gross-tax;}



int main()
{
float basic_salary;
printf("Enter Basic Salary:");
scanf("%f",&basic_salary);
float hra=calculateHRA(basic_salary);
float da=calculateDA(basic_salary);
float gross=calculateGross(basic_salary,hra,da);
float tax=calculateTax(gross);
printf("====================SALARY DETAILS====================\n");
printf("Gross Salary = %.2f\n",gross);
printf("Tax = %.2f\n",calculateTax(gross));
printf("Net Salary = %.2f\n",calculateNet(gross,tax));
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\week-6 submit> cd "c:\Users\Chetha
Enter Basic Salary:50000
====================SALARY DETAILS====================
Gross Salary = 65000.00
Tax = 3250.00
Net Salary = 61750.00
PS C:\Users\Chethan S\Documents\week-6 submit>
```

CODE

PROGRAM-2

c2.c > main()

```c
#include<stdio.h>

float calculateIncome(float reg_fees,float spons_amt,float stall_rent)
{return ((reg_fees*15)+spons_amt+stall_rent);}

float calculateVolunteerExpense(float No_of_volunteer)
{return No_of_volunteer*200;}

float calculateTotalExpense(float stage_setup,float guest_payment,float marketing_expense,float volunteer_expense)
{return stage_setup+guest_payment+marketing_expense+volunteer_expense;}

float ProfitOrLoss(float income,float expense)
{return income-expense;}

int main()
{
float reg_fees,spons,stall,stage,guest,marketing,volunteers,prof_or_loss;


printf("Registration :");
scanf("%f",&reg_fees);

printf("Sponsorship:");
scanf("%f",&spons);

printf("Stall Rent:");
scanf("%f",&stall);

printf("Stage setup:");
scanf("%f",&stage);

printf("Guest payment:");
scanf("%f",&guest);

printf("Marketing:");
scanf("%f",&marketing);

printf("Volunteers:");
scanf("%f",&volunteers);

float total_income=calculateIncome(reg_fees,spons,stall);
float volunteer_expense=calculateVolunteerExpense(volunteers);
float total_expenditure=calculateTotalExpense(stage,guest,marketing,volunteer_expense);

prof_or_loss=ProfitOrLoss(total_income,total_expenditure);



printf("Total income: %.2f\n",total_income);
printf("Total Expense: %.2f\n",total_expenditure);

if(prof_or_loss<0)
    printf("Loss: %.2f",prof_or_loss);
else
    printf("Profit: %.2f",prof_or_loss);

}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\week-6 submit> cd "c:\Users\Chethan S\Do
Registration :500
Sponsorship:20000
Stall Rent:10000
Stage setup:15000
Guest payment:8000
Marketing:5000
Volunteers:20
Total income: 37500.00
Total Expense: 32000.00
Profit: 5500.00
PS C:\Users\Chethan S\Documents\week-6 submit>
```

PROGRAM-3

CODE

```c
#include<stdio.h>

int avg_stock(int n,int array[n])
{   int sum=0;
    printf("Enter stock of %d books:\n",n);
    for(int i=0;i<n;i++)
    {
        scanf("%d",&array[i]);
    }
    int *ptr=array;
    for(int i=0;i<n;i++)
    {
        sum+=*ptr;
        ptr++;
    }

    return sum/n;

}


int main()
{
int n;
printf("Enter number of books:");

scanf("%d",&n);
printf("%d\n",n);
int books[n];

float avg=avg_stock(n,books);
printf("Average stock of books = %.2f", avg);
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\week-6 submit> cd "
Enter number of books:5
5
Enter stock of 5 books:
10 20 30 40 50
Average stock of books = 30.00
PS C:\Users\Chethan S\Documents\week-6 submit>
```

PROGRAM-4

CODE

```c
#include<stdio.h>

float total_fee_collection(int child[10])
{

    int total=0;
    for(int i=0;i<10;i++)
        total+=child[i];

    return total;

}

int total_salary_expenditure(int managers,int senior_caretaker,int assistant_caretaker)
{return managers*45000+senior_caretaker*30000+assistant_caretaker*15000;}

float total_additional_expense(float maintainance,float prizes)
{return maintainance+prizes;}

int main()
{
int manager,sen_care,assist_care;
float maintainance,prize;
int child[10];
printf("Enter fee paid by 10 children:\n");
for(int i=0;i<10;i++)
    {printf("Child %d Fee: ",i+1);
    scanf("%d",&child[i]);}

printf("Enter number of managers: ");
scanf("%d",&manager);

printf("Enter number of Senior Caretakers: ");
scanf("%d",&sen_care);

printf("Enter number of Assistant Caretakers: ");
scanf("%d",&assist_care);

printf("Enter maintenance: ");
scanf("%f",&maintainance);

printf("Enter total cash prizes for top performing children: ");
scanf("%f",&prize);
float total_fee=total_fee_collection(child);
float total_salary=total_salary_expenditure(manager,sen_care,assist_care);
float total_additional= total_additional_expense(maintainance,prize);
printf("============Play Area Budget Summary============\n");
printf("Total Fee collection = %.2f\n",total_fee);
printf("Total Salary Expenditure = %.2f\n",total_salary);
printf("Total additional expense = %.2f\n",total_additional);

float rem_balance=total_fee-(total_salary+total_additional);
printf("Final Remaining Balance = %.2f\n",rem_balance);

}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\week-6 submit> cd "c:\Users\Ch
Enter fee paid by 10 children:
Child 1 Fee: 20000
Child 2 Fee: 22000
Child 3 Fee: 21000
Child 4 Fee: 20000
Child 5 Fee: 23000
Child 6 Fee: 21000
Child 7 Fee: 22000
Child 8 Fee: 20000
Child 9 Fee: 21000
Child 10 Fee: 22000
Enter number of managers: 1
Enter number of Senior Caretakers: 1
Enter number of Assistant Caretakers: 3
Enter maintenance: 30000
Enter total cash prizes for top performing children: 15000
============Play Area Budget Summary============
Total Fee collection = 212000.00
Total Salary Expenditure = 120000.00
Total additional expense = 45000.00
Final Remaining Balance = 47000.00
PS C:\Users\Chethan S\Documents\week-6 submit>
```

CODE

PROGRAM-5

```c
#include <stdio.h>

int getRate(int type)
{
    if(type == 1)
        return 20;
    else if(type == 2)
        return 40;
    else if(type == 3)
        return 60;
    else
        return 0;
}

float calculateBill(int *hours, int rate, int n)
{
    float total = 0;
    int *ptr=hours;

    for(int i = 0; i < n; i++)
        total+= (*(ptr+i)) * rate;

    return total;
}

void  displayBills(int *hours, int *type, int n)
{
    float bill,totalRevenue = 0,max_bill = 0;
    int max_vehicle = 0;

    char *names[]={"Bike", "Car", "SUV"};

    printf("\n----- Parking Bill Report -----\n");

    for(int i = 0; i < n; i++)
    {
        int rate = getRate(type[i]);
        bill = hours[i] * rate;

        printf("Vehicle %d (%s): Hours = %d, Bill = %.2f\n",i+1, names[type[i]-1], hours[i], bill);

        totalRevenue += bill;

        if(bill > max_bill)
        {
            max_bill = bill;
            max_vehicle = i;
        }
    }
    printf("Total Revenue = %.2f\n", totalRevenue);
    printf("Maximum Bill Paid By Vehicle %d (%s) = %.2f\n",max_vehicle+1, names[type[max_vehicle]-1], max_bill);
}
```

```c
int main()
{
    int n, threshold, count = 0;
    printf("Enter number of vehicles: ");
    scanf("%d", &n);

    int hours[n], type[n];

    printf("Enter details of each vehicle:\n");
    for(int i = 0; i < n; i++)
    {
        printf("Vehicle %d type: ", i+1);
        scanf("%d", &type[i]);

        printf("Vehicle %d hours: ", i+1);
        scanf("%d", &hours[i]);
    }
    int *ptr[n];

    for(int i = 0; i < n; i++)
        ptr[i] = &hours[i];


    printf("Enter threshold hours: ");
    scanf("%d",&threshold);

    displayBills(hours, type, n);
    for(int i = 0; i < n; i++)
    {
        if(*ptr[i] > threshold)
            count++;

    }
    printf("Vehicles parked more than %d hours = %d\n", threshold, count);
    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\week-6 submit> cd "c:\Users\
Enter number of vehicles: 5
Enter details of each vehicle:
Vehicle 1 type: 2
Vehicle 1 hours: 3
Vehicle 2 type: 1
Vehicle 2 hours: 6
Vehicle 3 type: 3
Vehicle 3 hours: 2
Vehicle 4 type: 2
Vehicle 4 hours: 8
Vehicle 5 type: 1
Vehicle 5 hours: 1
Enter threshold hours: 5

----- Parking Bill Report -----
Vehicle 1 (Car): Hours = 3, Bill = 120.00
Vehicle 2 (Bike): Hours = 6, Bill = 120.00
Vehicle 3 (SUV): Hours = 2, Bill = 120.00
Vehicle 4 (Car): Hours = 8, Bill = 320.00
Vehicle 5 (Bike): Hours = 1, Bill = 20.00
Total Revenue = 700.00
Maximum Bill Paid By Vehicle 4 (Car) = 320.00
Vehicles parked more than 5 hours = 2
PS C:\Users\Chethan S\Documents\week-6 submit>
```